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Tracking Europe's AI transition: A new STOXX 600 earnings-call framework

November 11, 2026

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How quickly is AI moving from experimentation into everyday business, and where are companies reporting tangible gains? The answers matter for institutional investors and policymakers assessing productivity, operating efficiency and labour demand. Yet comparable evidence across Europe remains limited, making it difficult to track the pace and scale of the transition. [1]

We address this gap with a proprietary framework that turns what is said on STOXX 600 earnings call into three structured indicators: AI adoption intensity, reported productivity benefits, and the impact on workforces. Large language models analyse what management teams say about AI's operational role, applying a common scoring framework across companies, sectors and countries. Our new approach provides a timely pan-European comparable tracker that captures not just whether AI is being mentioned, but its real-world impact as the adoption of the new technology unfolds.

Key Takeaways:

- **Adoption is broadening, with signs of faster progress.** AI-use scores rose gradually from a low base before strengthening over the past 12 months. The increase extends across sectors, although the pace and level of adoption remain uneven.
- **Workforce impacts remain limited in corporate disclosures.** The framework does not show a broad negative workforce signal attributable to AI. The aggregate score remains close to the level of no impact on the labour force across all of our sample. However, this does not rule out AI displacing workers at a future date.
- **Geographic differences persist within the listed-company sample.** Since 2022, German companies in the STOXX 600 have generally recorded above-average AI-use and reported productivity scores. Dutch and Spanish companies in the STOXX 600 have tended to score lower on AI use, although Spain has recently gone through a rapid catch up.
- **The technology J-Curve:** The findings are consistent with an early stage of the AI transition. Companies report increasing adoption of the new technology, but transformational productivity gains and extensive workforce restructuring remain uncommon.

[1] 1 Our US team regularly publish on data from the USCB showing the impact of AI on businesses, and our German team have recently begun their own chartbook. In the UK, the Office for National Statistics publishes on the topic too.

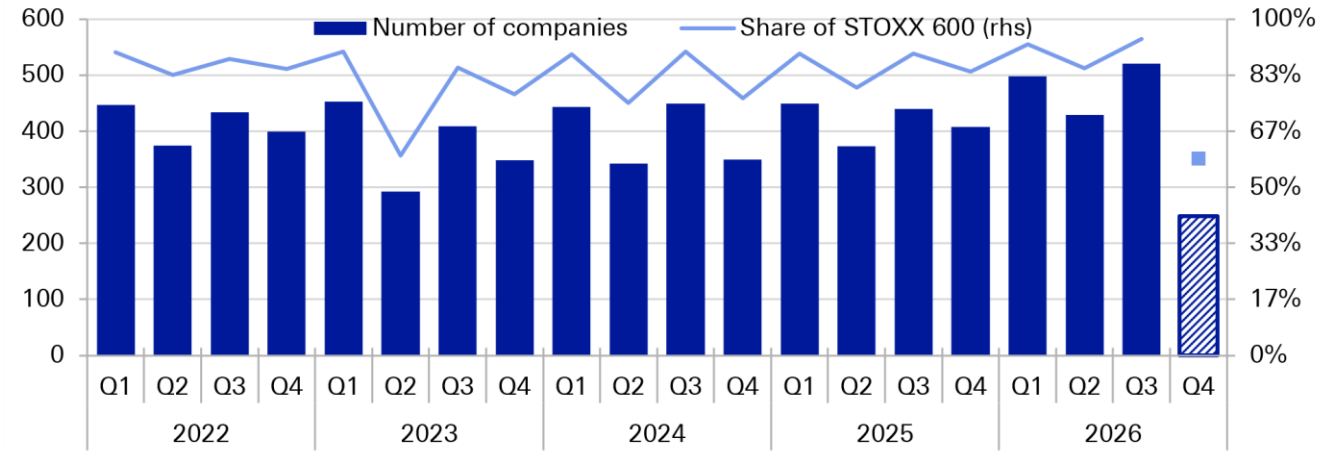
Methodology

Sample period: Our sample begins in January 2022, providing a baseline before ChatGPT's launch on 30 November 2022. AI in this framework covers both established and emerging technologies, so the launch of ChatGPT is a reference point rather than the start of all corporate AI adoption.

Timing of observations: STOXX 600 companies report on different financial calendars and at quarterly or semi-annual intervals. Their earnings calls also mix discussion of completed activity with forward-looking guidance.

We assign each transcript to the calendar quarter in which the earnings call takes place, regardless of the financial reporting period discussed. The resulting indicators therefore track when information about AI use and its operational impacts becomes public rather than when implementation or the reported effect occurred. This provides a common dating convention across the sample while retaining differences in reporting frequency and disclosure timing.

Figure 1: Coverage of the STOXX 600 earnings calls



Source: Deutsche Bank Research. Notes: Share of the STOXX 600 is based on the current company weight in the index. Data collection for the current quarter (Q4 2026) is incomplete as some companies are still to report

Constructing the STOXX 600 AI corpus: We use a specialised AI transcription engine to process earnings calls from STOXX 600 companies, restricting the analysis to calls for which we obtain complete transcripts of sufficient accuracy. In Q3 2026, the latest full quarter, the dataset covered more than 500 companies, representing approximately 90% of the index's current market capitalisation weights (Figure 1). In the current quarter (Q4) earnings calls are still taking place. Nearly 250 companies with an equivalent weight to around c.60% of the STOXX index having already reported.

Historical coverage typically ranges from 300 to 500 companies and represents more than 80% of the index's current market-capitalisation weights. Coverage peaks twice a year because some European companies report semi-annually.



Algorithmic Scoring

Each transcript is analysed by an LLM using a structured prompt to classify disclosures across three dimensions: AI use, reported productivity impact and workforce impact. The LLM approach bypasses simple keyword counts to evaluate semantic context, distinguishing between casual mentions and concrete operational disclosures. A common scoring framework enables comparisons across the covered companies.

- **AI Use Score** summarises the intensity of AI adoption described in a company's earnings call on a scale from 0 to 10. It distinguishes limited experimentation from operational implementation and AI-centred business models:
    - **0:** No mention of AI, machine learning, or automated algorithmic systems.
    - **1–3 (Low):** Minor mention of AI (e.g., pilot programs, exploring ideas, client-side adoption only).
    - **4–7 (Medium):** Active implementation (e.g., AI is integrated into specific operational workflows, products, or core services).
    - **8–10 (High):** Core business driver (e.g., business model is centred on AI, massive R&D spend dedicated to AI, transformational integration).
- **AI Productivity Impact** score summarises company disclosures about realised productivity and profit benefits attributed to AI on a scale from 0 to 10:
    - **0:** No productivity impact mentioned, or AI is not used.
    - **1–3 (Low):** Minor efficiencies (e.g., slightly faster administrative tasks).
    - **4–7 (Medium):** Notable operational improvements (e.g., significant cost savings or volume increases in specific departments).
    - **8–10 (High):** Transformational productivity gains (e.g., doubling output, rendering legacy processes obsolete).
- **AI Labour Force Impact** score summarises disclosures about AI-related changes in labour demand and workforce development on a scale from −5 to +5. Negative scores reflect workforce reductions or hiring restraint; positive scores include hiring and the creation of AI-focused divisions:
    - **-5:** Significant workforce reduction or hiring freezes directly attributed to AI automation.
    - **0:** No impact on headcount or hiring decisions, or AI labour impact is not mentioned.
    - **+5:** Aggressive hiring or creation of new divisions dedicated to AI.

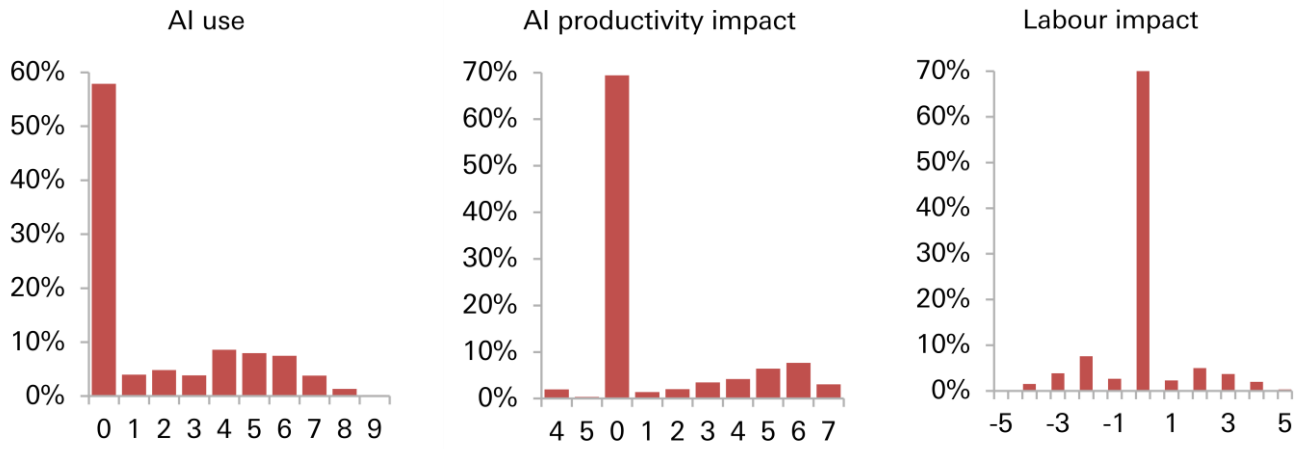
For examples of the AI's scoring and justifications, see the tables presented in the Annex at the end of this report.

Summary statistics

Figure 2 shows the distribution of scores across all earnings-call transcripts in the historical sample, without weighting for company size or reporting frequency. These are transcript-level distributions rather than shares of companies, and they combine early and recent observations. They provide context for the quarterly indicators:

- **AI use score:** Approximately 60% of earnings call transcripts do not discuss AI use. Among the minority that do discuss it, most fall into the "Active implementation" range (a score of 4-6 out of 10).
- **AI's productivity impact:** Nearly 70% of earnings calls do not mention a tangible AI-related productivity benefit. Among calls that do, scores are generally in the medium-impact band, indicating notable operational improvements. Transformational gains, scored at 8 or above out of 10, are rarely reported.
- **AI's labour impact:** Only around 10% of earnings calls discuss AI-related workforce adjustments. Reported changes typically receive scores of 1–2 in magnitude on the –5 to +5 scale, with a slight tilt towards hiring and upskilling rather than redundancies.

Figure 2: Raw distribution of scores across the entire sample



Source: Deutsche Bank Research. Note: Results are based on the raw number of transcripts, and are independent of firm size, and biased towards firms reporting more often.

Taken together, the distributions are consistent with an early adoption phase: companies discuss AI implementation more often than transformational productivity gains or extensive workforce restructuring. **A technology J-curve offers one interpretation of this pattern,** with investment and organisational adaptation preceding broader benefits.

This J-curve pattern is strongly corroborated by external micro-data, such as the Banco de España's business activity survey (EBAE) of Spanish firms, which

similarly indicates that AI has a more immediate impact on upfront investment and productivity than on headcount.

AI use: A broad-based rise, with signs of faster recent adoption

The new AI-use indicator points to a broadening transition across STOXX 600 companies, with early signs that adoption has accelerated recently. Scores rose only gradually from 2022 and 2024, but the latest observations suggest a stronger pace across industries and country groups.

The rise is broad-based rather than confined to one sector. Technology has led much of the sample, while financials show tentative recent outperformance and utilities have caught up from a low starting point. Among the large country groups, German companies have persistently recorded above-average scores.

AI use across European industries

To trace sectoral dispersion, we group companies by STOXX industry classification and calculate a weighted average of their scores for each calendar quarter, using their latest STOXX 600 index weights.[2]

We focus on four sectors spanning different business models: financials, the largest services sector; industrials, the largest physical-assets sector; technology, a likely early adopter; and utilities, an infrastructure-heavy sector where adoption may proceed more slowly.

Figure 3 presents the AI use score for these four key industries alongside the range for all industry scores (represented by the shaded area).

- The AI use score is generally low over time but is gradually rising from around 1 point in 2022 to around 3 in 2026.
- AI-use scores are rising across industries, but differences between sectors are widening. The common direction therefore points to broadening adoption rather than uniform progress: sectors can participate in the transition while remaining at different levels of use.

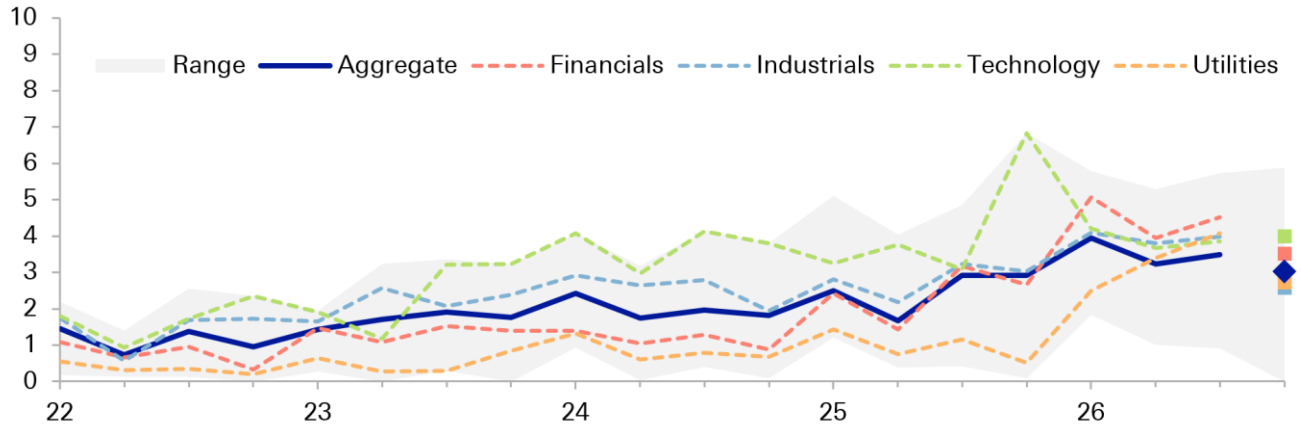
In terms of the industries of interest:

- **Technology's early lead has narrowed.** The sector's AI-use score was near the top of the industry range through to 2024 but has recently plateaued as the broader STOXX 600 average caught up during 2025–26.[3]
- **Financials and industrials broadly track the index average.** Both sectors have remained close to the aggregate AI-use score over history, with financials generally a little below average, and industrials a little above. But financials show tentative signs of moving above average in recent quarters.

[2] The industries are: (1) Basic Materials, (2), Consumer Discretionary, (3), Consumer Staples, (4) Energy, (5) Financials, (6) Health Care, (7) Industrials, (9) Real Estate, (10) Technology, (11) Telecommunications, (12) Utilities.
[3] Q4 2025 saw a large increase in the AI use score for the "Technology" industry, jumping up to around 6. This one-off spike was driven by ASML's announcement of a significant strategic partnership with and investment in Mistral AI" to 'embed AI across our entire holistic portfolio in order to increase the performance and productivity of our systems and the yield of our customers' processes'. Because of ASML's large market cap (around 49% of our Technology category), it distorts the weighted average significantly.

- **Utilities have caught up from a low base.** After recording lower AI-use scores than other sectors for much of the sample, utilities converged towards the STOXX 600 average during 2026. The result shows that the rise in reported adoption extends beyond the sectors most closely associated with AI.

Figure 3: AI use score (0-10) by industry



Source: Deutsche Bank Research. Notes: The value for each industry is computed as a weighted average of all firms within the industry where the weights are derived from the current STOXX 600 share of each company. The markers denote the readings for the current incomplete quarter

As a cross-check, our findings align with the European Commission's October Eurobarometer Flash survey of MSEs and Mid-Caps and the Eurostat annual ICT surveys, which show low overall intensity but a similar expansion in service- and tech-related sectors.

AI use across countries

Companies in the STOXX 600 tend to be large multi-nationals. That makes it difficult to pin down where physically the AI transformation is taking place within their organisation. Instead, our country comparison is done using two classifications of a company's country:

- **Country of domicile:** AI is a transformation technology, and this type of change is likely to be driven in organisation from the top down. We therefore consider a company's country as that in which it is domiciled (which is likely to be where the headquarters is also).
- **Country of primary listing:** While companies can generally list in other countries, there tends to be a home bias in their choice of exchange.

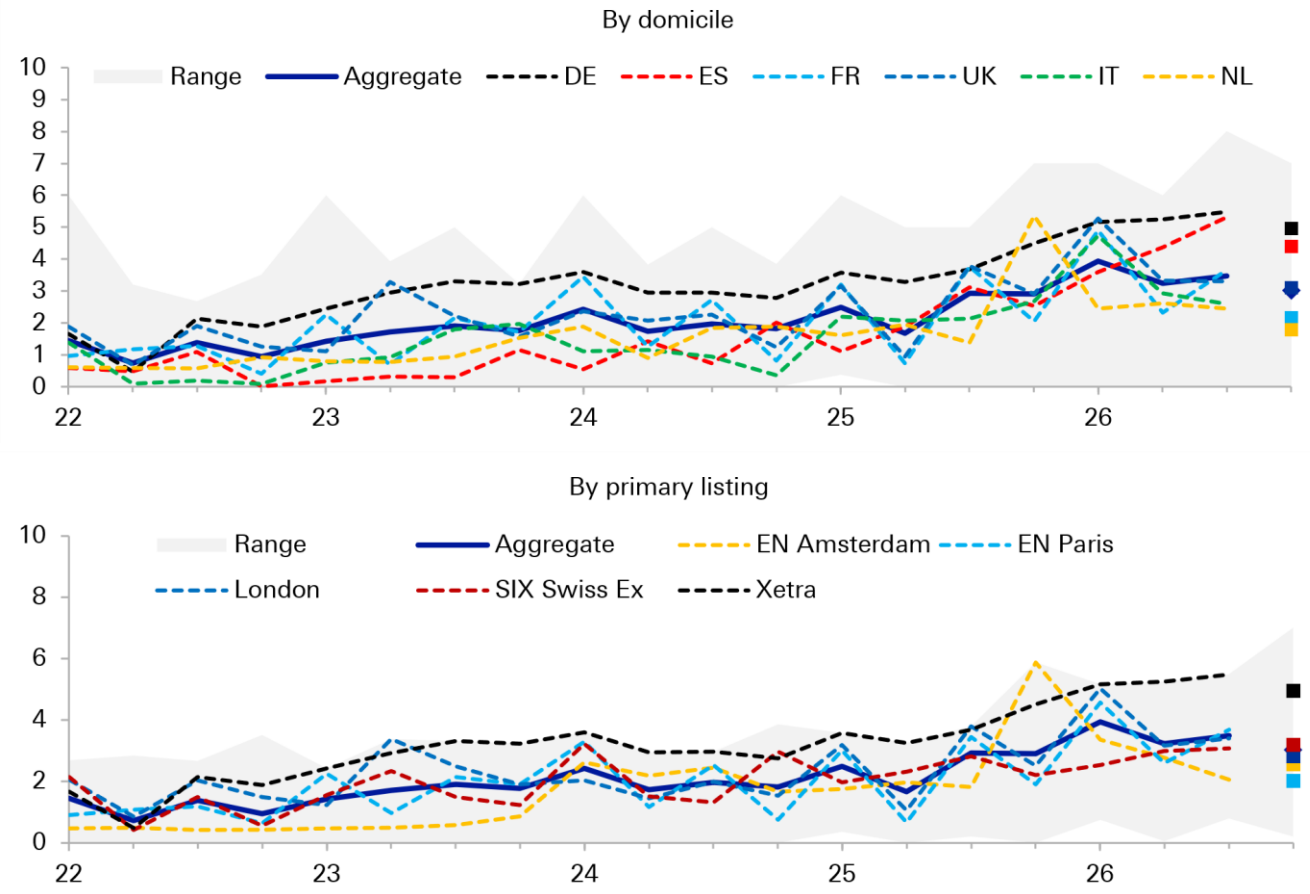
Figure 4 presents the AI use intensity metric by both of these measures of country (the company's domicile and primary exchange listing). Again, the shaded area shows the range across all classifications, with the largest countries/exchanges singled out.

German-domiciled companies and the Xetra-listed (Frankfurt stock exchange) group record above-average AI-use scores in most quarters since 2022.

Spain's AI-use score has risen noticeably over recent quarters and now exceeds the aggregate average, reversing its earlier below-average position.

AI-use scores rise across country groups, with indications of faster growth from 2025. The shared direction suggests a broad European listed-company trend, even though differences in score levels and timing persist between countries.

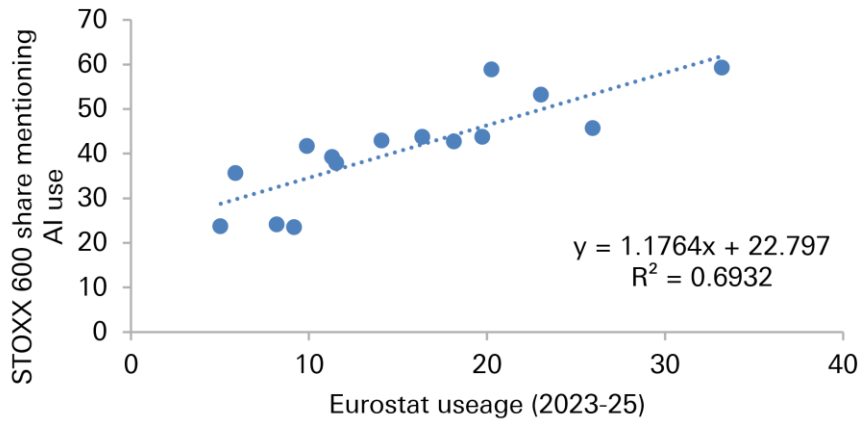
Figure 4: AI use intensity (0-10) by country



Source: Deutsche Bank Research. Notes: The markers denote the readings for the current incomplete quarter

As a geographical cross-check, our index matches the broad trends of the Eurostat ICT annual survey, which places Germany as a clear frontrunner among its large peers. But some differences do exist. For instance, Eurostat shows the Netherlands with a very high AI adoption. As Figure 5 demonstrates, there is a positive correlation between the AI use score metric derived here, and that from Eurostat, even though the Eurostat data measures if firms are adopting AI, and our new use measure also incorporates the concept of how intensively firms are using AI.

Figure 5: Relationship between Eurostat usage and Weighted share of STOXX 600 mentioning AI usage



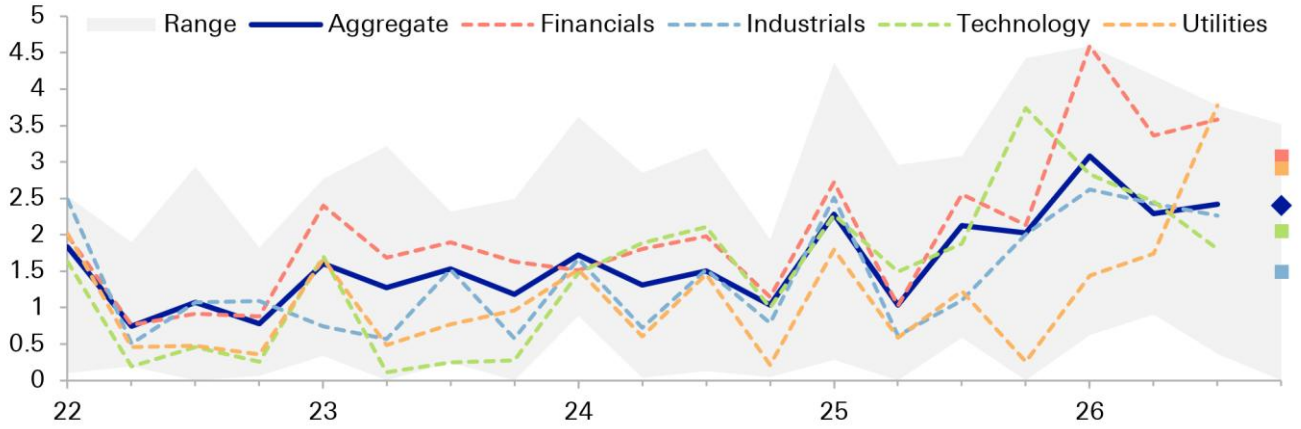
Source: Deutsche Bank Research, Eurostat. Notes: each dot represents a country-year pair of AI usage>0

AI productivity impact: Reported benefits are rising, but transformational gains remain uncommon

Higher intensity of use does not necessarily mean firms are benefiting more from AI. But our AI productivity impact metric does show a similar pattern to our AI usage metric, although it tends to be slightly lower, indicating that companies may not yet be reaping the full rewards for their AI investment.

Financials stand out more clearly on reported productivity benefits than on AI-use intensity, while German companies again record relatively strong scores, but only in more recent history.

Figure 6: AI's impact on productivity (0-10) by industry



Source: Deutsche Bank Research. Notes: The markers denote the readings for the current incomplete quarter

AI productivity impact across European industries

Reported AI-productivity benefits have become more prominent across industries. Figure 6 shows scores rising gradually during 2022–25, followed by

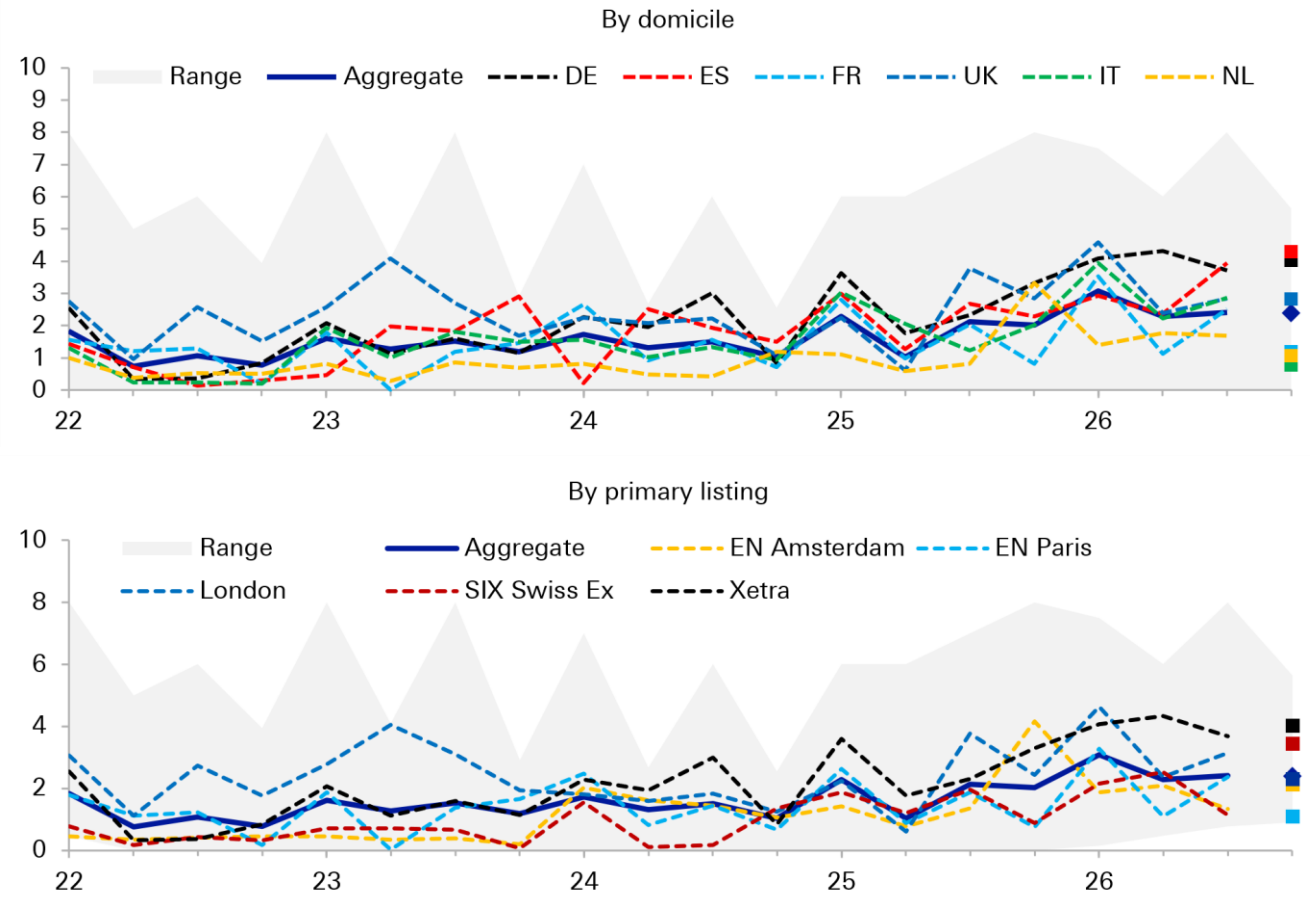
indications of a faster increase from late 2025. The profile resembles the AI-use trend in Figure 4, suggesting that the stronger adoption narrative is accompanied by more discussion of operational gains.

The aggregate trajectories of the AI use and AI productivity impact scores are similar, but sector rankings differ. Technology's early relative strength in AI use is not matched by its productivity score, while financials report above-average productivity benefits despite more modest use scores.

AI productivity impact across countries

Figure 7 plots the self-report impact of AI on company productivity by country. Between 2022-23, German companies recorded average productivity impact, even though their AI use score was relatively high. However more recently, their AI impact score has persistently been above average. The UK grouping shows persistent early outperformance in AI impact during 2022–23, followed by convergence towards the aggregate average.

Figure 7: AI's impact on productivity (0-10) by country

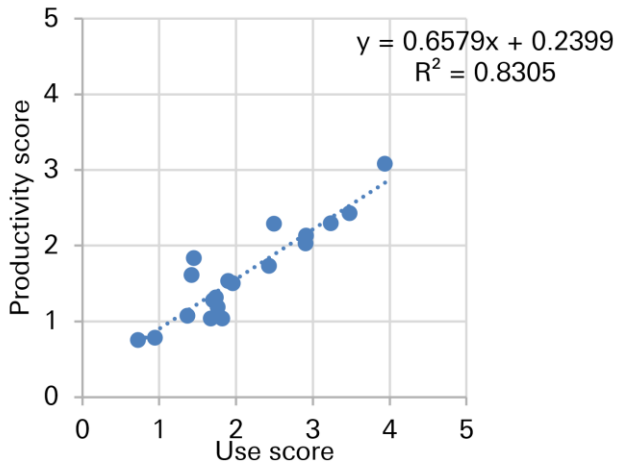


Source: Deutsche Bank Research. Notes: The markers denote the readings for the current incomplete quarter

Figure 8 shows that there is a strong positive correlation between the AI use score and the AI productivity score across the STOXX 600 (r-squared = 0.83). The relationship between the scores is approximately linear, with a slope of around

0.66: a one-point higher use score is associated with roughly a 0.66-point higher productivity score for each country-year.

Figure 8: Productivity and use scores are correlated, but less than 1:1

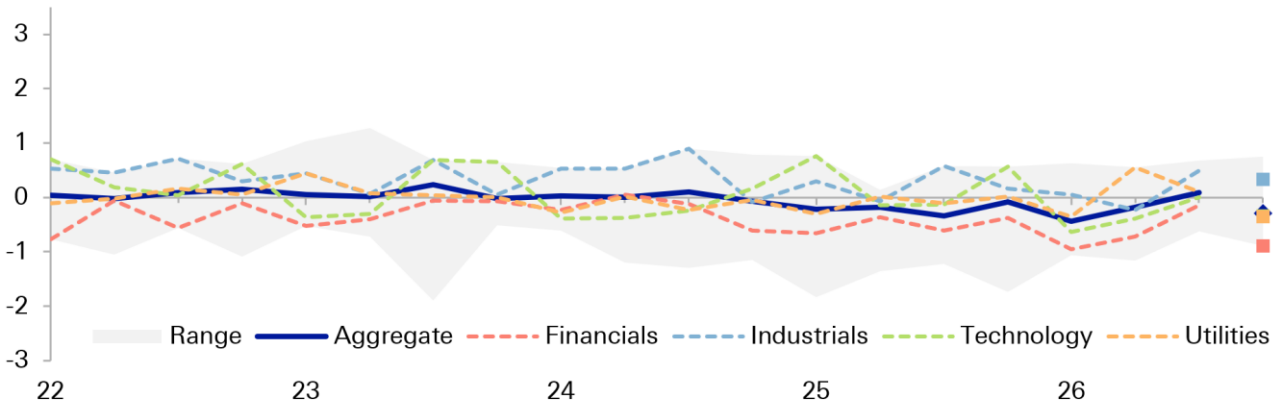


Source: Deutsche Bank Research

AI labour force impact: On net, AI may be benefiting employment, for now...

Across industries and countries, the recorded score for AI's impact on the labour force remains relatively small. On average, the reported impact on companies' employment has been fairly neutral, with a marginal negative bias over the last year or so.

Figure 9: AI's impact on the labour force (-5 to +5) by industry



Source: Deutsche Bank Research. Notes: The markers denote the readings for the current incomplete quarter. Negative (positive) scores are associated with job losses (gains) associated with a company's AI use.

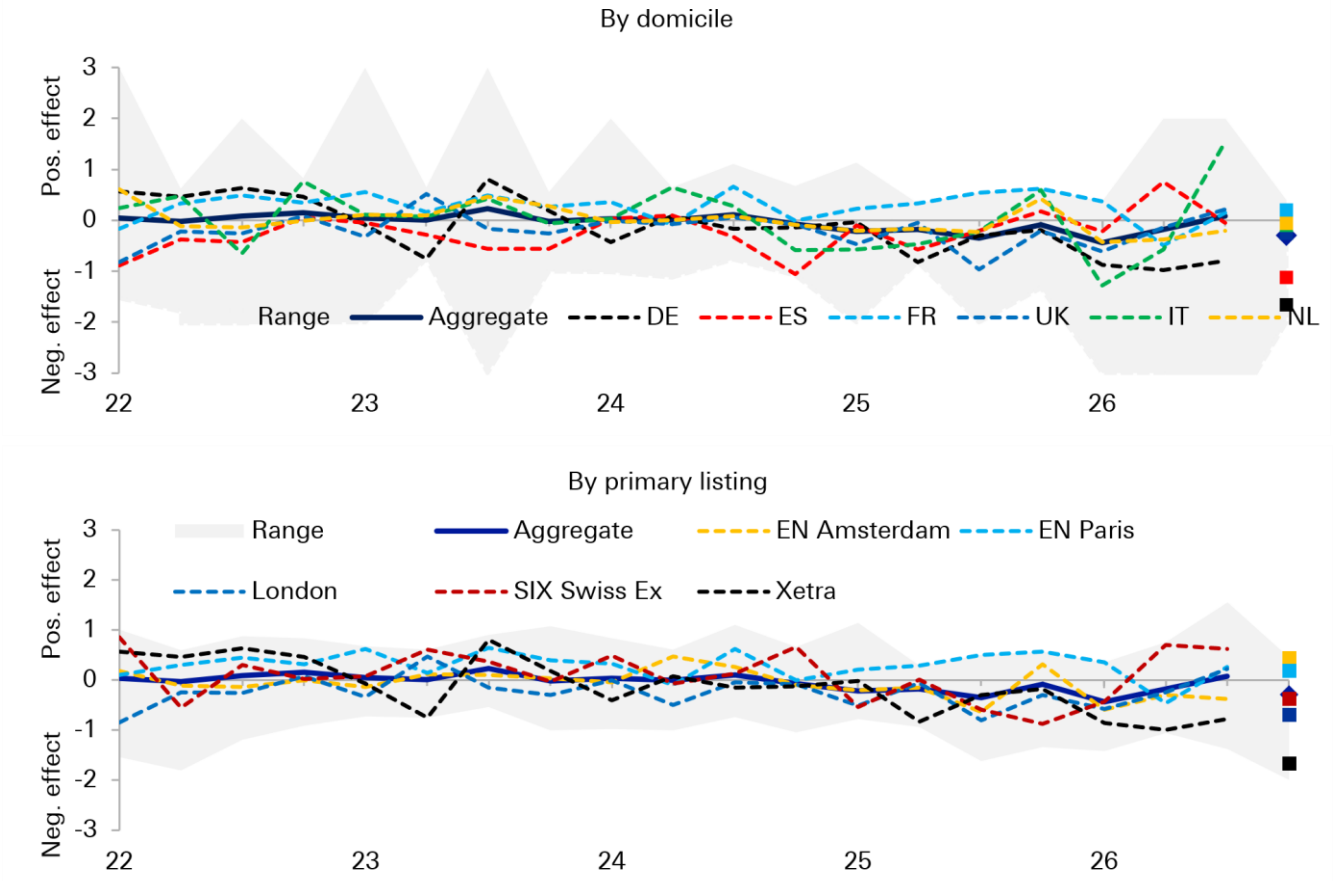
AI labour force impact across European industries

Despite rising productivity, AI has not triggered the negative employment shock feared by the public. Aggregate labour impact scores across nearly all sectors hover near zero, with a slightly positive balance over history, and a slightly negative balance more recently in 2025-26 (Figure 9). But overall the magnitude of AI's impact on labour remains low. For the STOXX 600 aggregate average, the score's maximum deviation from 0 (no impact) is 0.43 on our 0-5 scale.

AI labour force impact across countries

Workforce-impact scores are also small across the main country groupings, with most countries slowing a slight negative tilt in recent years (Figure 10).

Figure 10: AI impact on the labour force (-5 to +5) by country



Source: Deutsche Bank Research. Notes: The markers denote the readings for the current incomplete quarter. Negative (positive) scores are associated with job losses (gains) associated with a company's AI use.

The lack of significant impact so far from AI on companies' labour forces so far is consistent with results from our previous work looking at the other side of the labour market. Data from our dbDataInsights household survey showed that workers across Europe reported a fairly low level of fear about losing their job to AI over the next year. The analysis in that report found results that were more consistent with workers adapting to the new AI world by upskilling and



demanding higher wages. Those results would not be consistent with a world in which AI was actively replacing labour.

These findings of a low impact on labour are highly consistent with recent academic research (e.g., Yotzov et al., 2026), which found that nearly half of surveyed UK and German firms reported absolutely no impact on headcount from AI over the past three years,

Conclusion

The newly developed framework provides a recurring way to monitor company disclosures about AI's impact on productivity and labour. **The central finding of this report is a broadening AI transition rather than clear evidence of an economy-wide productivity breakthrough.** The new metrics indicate that AI use remains limited but is rising across the European countries and industries represented in the sample. Companies report low but growing productivity gains and little aggregate workforce impact so far. This pattern is consistent with a technology J-curve, with firms still integrating the new technology, rather than proof that this trajectory will unfold. Limited reported workforce effects today do not rule out subsequent layoffs or wider organisational restructuring.

Quarterly updates to this framework will track Europe's AI transition and provide timely signals of whether companies are reporting more substantial productivity and employment effects.

Beyond AI, the framework could be adapted to extract signals about employment decisions, investment decisions and the general economic outlook from earnings calls. These potential extensions could form the basis of alternative nowcasting models for important macroeconomic data; they are applications that will be explored in future work.

Annex

The tables below provide examples of the LLM's scoring of earnings call transcripts for the various AI metrics developed in this report. Some examples have been shortened for brevity in the tables, and names of companies, people, etc have been redacted.

Figure 11: AI adoption intensity

Scoring matrix	Examples
0: No mention of AI, machine learning, or automated algorithmic systems.	There is no mention of Artificial Intelligence (AI), machine learning, or any related automated systems in the provided transcript.
1–3 (Low): Minor mention of AI (e.g., pilot programs, exploring ideas, client-side adoption only).	**Score = 2:** The company's discussion of AI is minimal and primarily future-oriented, focused on investment rather than current internal adoption. The main mention of AI comes when CEO [NAME] discusses a new EUR 2 billion venture capital fund, stating, 'The areas we wanna invest into is network security, IoT, GenAI, and some opportunities for beyond growth activities in the core business.' However, he clarifies that the fund is still being set up and has not made any investments yet. This indicates an exploratory or planning stage.
4–7 (Medium): Active implementation (e.g., AI is integrated into specific operational workflows, products, or core services).	**Score=5:** The primary example is the recent acquisition of [COMPANY], a company with IoT solutions and 'software that interprets all of those signals' from vibration sensors to 'predict from that data what's gonna go wrong'. This technology is being integrated into the core Process Automation division to accelerate aftermarket growth. Supporting this, the company also highlights developing 'intelligent' products like the [PRODUCT], and using data analytics in its CRM to 'find patterns' for commercial decisions. This indicates an active implementation of AI-related technologies into specific products and services, warranting a medium score.
8–10 (High): Core business driver (e.g., business model is centred on AI, massive R&D spend dedicated to AI, transformational integration).	**Score=8:** ... The CEO, [NAME], dedicates a significant portion of the call to AI, calling it a "once-in-a-generation opportunity" where the firm intends to "lead the charge." The adoption is not theoretical; the company reports that an exceptionally high "96% of the firm already using AI tools on a daily basis." This is supported by concrete examples of active, value-additive implementation, such as a discretionary portfolio manager using a "proprietary AI toolkit" to double their coverage universe. The company is also building its own "cutting-edge AI infrastructure" and connecting advanced models to its proprietary data. The strategy is to "reshape the entire firm," applying AI as a "multiplier" across both investment functions (quant research) and operations (client onboarding). This goes far beyond pilot programs and represents a deep, strategic, and transformational integration of AI into the core business.

Source: Deutsche Bank Research

Figure 12: AI's impact on company productivity

Scoring matrix	Examples
0: No productivity impact mentioned, or AI is not used.	The transcript does not contain any mention of AI or its impact on the company's internal productivity or efficiency.
1–3 (Low): Minor efficiencies (e.g., slightly faster administrative tasks)	**Score=2**: The company's Vice President, [NAME], made a very brief mention of AI in his closing remarks. He stated, "Just to put an end, talk just a second on the prudent deployment of AI and digitalization that is going on in the company. The company is improving a lot... we are expanding it over the company." While this confirms the internal use of AI and claims positive impact ("improving a lot"), the statement is extremely high-level and lacks any specific examples, metrics, or details about productivity gains or cost savings. The description of a "prudent deployment" suggests a cautious, incremental approach. The brevity and lack of substance indicate that the impact to date is likely minor.
4–7 (Medium): Notable operational improvements (e.g., significant cost savings or volume increases in specific departments).	**Score=5:** The company explicitly states it is using AI to improve internal productivity, particularly within its large technology division. In response to a question about productivity, CFO [NAME] noted, 'we've got something like 11,000 employees involved in technology. And we're becoming more productive as we're able to use better tooling, including AI tooling, in the product development lifecycle.' This productivity gain is cited as a component of the company's 'strong year of organic margin progression.' While many AI mentions relate to customer-facing products, this specific point confirms a notable operational improvement within a core internal function, justifying a score in the medium range.
8–10 (High): Transformational productivity gains (e.g., doubling output, rendering legacy processes obsolete).	**Score=8:** The company explicitly details a significant productivity impact from its use of artificial intelligence. CEO [NAME] stated that through AI-powered value stream mapping, they were able to dramatically reduce production throughput time at their Malta site from 50 days down to 25 days. This 50% reduction in a core operational process represents a transformational gain, effectively doubling the speed and potential output of that production line. The CEO described this as 'top class' and noted it also helps release cash and improves responsiveness to customer demand. The success has prompted a plan to implement this AI-driven approach across the entire [COMPANY] network, indicating the high value and impact of this initiative.

Source: Deutsche Bank Research

Figure 13: AI's impact on the labour force

Scoring matrix	Examples
Negative scores (-5 to -1)	**Score=-4:** The company's CEO, [NAME], explicitly links a workforce reduction to the company's transformation strategy involving automation. When discussing the UK market, he states, "we are actually lowering the amount of people that we have. We started the year with 20,000 people. We're going to end up with 18,000. So simplification and automation is well underway." This represents a 2,000-person (10%) reduction in that geography directly attributed to automation... **Score=-2:** CEO [NAME] stated, "We've also launched a new machine learning backed version of contact form that now enables customers to resolve the problems within seconds using bots in over 40% of cases." Automating 40% of this workflow directly reduces the demand for human labor in that specific area. This aligns with the company's broader [INITIATIVE NAME] initiative and the CFO's comment that hiring is slowing down: "We've only added 4.6% of new staff during the last 6 months and it continues to slow." While no layoffs were announced, the implementation of labor-saving AI at this scale points to a clear negative impact on the company's demand for workers.
Positive scores (+1 to +5)	**Score=3:** CEO [NAME] explicitly stated that the company's recruitment efforts, which are 'even slightly ahead of the plan,' are 'instrumental... for the acquisition of new competencies to support our ambitions in digitalization, automation, and decarbonization.' This directly links an increase in labor demand (hiring) to the company's adoption of AI-type technologies. There was no mention of AI or automation leading to workforce reductions; the entire focus was on hiring to build capabilities in these areas. **Score=5:** CEO [NAME] announced a EUR 2 billion investment in generative AI over the next three years, a significant portion of which is allocated to personnel. He stated, "we intend to double our data and AI teams to 60,000 within the next three years." This represents a plan to hire 30,000 new employees for their data and AI teams. While the company's overall headcount has slightly decreased, this was attributed to market softness and optimizing operational buffers, not AI-driven automation. The clear and substantial plan to double the AI-specific workforce is a strong indicator of AI increasing the company's demand for labor.

Source: Deutsche Bank Research

Appendix 1

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